Answer key: Lesson 7, Take Away, Count Up

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Here is 68 as bars and squares. Cross out 2 bars. Fill in what is left.

Answer: 4 tens and 8 ones are left; 68 - 20 = 48

68 is 6 tens and 8 ones. Take away 2 tens: 4 tens and 8 ones are left. That is 48. The ones did not change.

Watch for: Crossing out 2 squares instead of 2 bars (66).; Writing 84 or 4 + 8 = 12.

S2. Take away. Use your kit if you like.

Answer: 37, 53, 56, 81

(a) 5 tens take away 2 tens is 3 tens: 37. (b) 7 ones take away 4 ones is 3 ones: 53. (c) 8 tens take away 3 tens is 5 tens: 56. (d) 6 ones take away 5 is 1: 81.

Watch for: Changing the wrong digit (57 - 20 = 55).; Taking away from both digits.

S3. Take away the tens first, then the ones. Find 68 - 25.

Answer: 48, then 43

68 take away 20 is 48. Then 48 take away 5 is 43. So 68 - 25 = 43.

Watch for: Taking away 25 as 2 + 5 = 7 (61).; Forgetting to start part (b) from 48.

S4. Take away. Then check by adding the answer back.

Answer: 34, then 46 (the same as the start, so it checks)

46 take away 10 is 36, and 36 take away 2 is 34. Check: 34 + 12 = 46, the number we started with. So 46 - 12 = 34 is right.

Watch for: Not adding back the number taken away (adding 46 + 12).; Writing 58 for the check.

S5. Mei has 45 stickers. She gives 13 away. Draw 45 as bars and squares. Cross out 13. How many are left?

Answer: 32 (4 bars and 5 squares drawn; 1 bar and 3 squares crossed out)

45 is 4 tens and 5 ones. 13 is 1 ten and 3 ones. 45 take away 10 is 35. 35 take away 3 is 32. So Mei has 32 left.

Watch for: Crossing out 13 squares.; Writing 58 (adding instead of taking away).

S6. Siti says 59 - 20 = 57. Is she right? Write the right answer, and say what she did.

Answer: No. 39. She took away 2 ones instead of 2 tens (57 is 59 - 2).

59 is 5 tens and 9 ones. Take away 2 tens: 3 tens and 9 ones, which is 39. Siti took away 2 ones, so she got 57.

Watch for: Saying she is right.; Writing 79 (adding 20).

S7. A bus has 64 seats. 21 seats have a child on them. How many seats are empty?

Answer: 43 seats

64 take away 20 is 44. Then 44 take away 1 is 43. So 43 seats are empty.

Watch for: Adding 64 + 21 = 85.; Taking away only 2 or only 1.

Think harder (page 3)

T1. Mei has 62 beads. Ravi has 38 beads. How many more beads does Mei have? Count up from 38: to the next ten, then by tens, then to 62.

Answer: 2, 20, 2, so 24 more beads

38 to 40 is 2. 40 to 60 is 20. 60 to 62 is 2. The whole gap is 2 + 20 + 2 = 24. Check: 38 + 24 = 62.

Watch for: Forgetting that two tens is 20 (writing 2).; Writing 62 - 38 as 36 or 34.

T2. Find 83 - 77 by counting up from 77.

Answer: 3, 3, so 83 - 77 = 6

77 to 80 is 3. 80 to 83 is 3. The gap is 3 + 3 = 6. Check: 77 + 6 = 83.

Watch for: Writing only one hop (3).; Taking away 83 - 7 = 76.

T3. Ali has 24 stickers. Ben has 31 stickers. Each child gets 10 more stickers. Now how many more stickers does Ben have than Ali?

Answer: 7 more (the gap stays the same: 34 and 41, and 41 - 34 = 7)

Ali now has 34 and Ben has 41. The gap from 34 to 41 is 7. It was 7 before too (24 to 31), because both children got the same 10 more.

Watch for: Writing 17 (7 + 10).; Writing 41 (Ben's number).

T4. A book has 71 pages. Ravi has read 46 pages. How many pages are left to read? Count up from 46.

Answer: 25 pages (46 to 50 is 4, 50 to 70 is 20, 70 to 71 is 1)

46 to 50 is 4. 50 to 70 is 20. 70 to 71 is 1. The gap is 4 + 20 + 1 = 25. Check: 46 + 25 = 71.

Watch for: Adding 71 + 46.; Forgetting the last hop of 1 (24).

Go further (page 4)

G1. Ravi wants to find 71 - 68 and 71 - 20. Which tool is quicker for each, take away or count up? Write why. Then find both answers.

Answer: 71 - 68 = 3 (count up: the numbers are close). 71 - 20 = 51 (take away: 2 tens).

71 - 68: count up. 68 to 70 is 2 and 70 to 71 is 1, so 3. The numbers are close, so the gap is short. 71 - 20: take away 2 tens to get 5 tens and 1 one, which is 51. Counting up 20 would be a long way.

Watch for: Taking away 68 from 71 one at a time.; Counting up from 20 to 71 (too long).

G2. A crate holds 100 eggs. 64 eggs are sold. How many eggs are left? Find it two ways, take away and count up. Do the two answers match?

Answer: 36 eggs. Take away: 100 - 60 = 40, then 40 - 4 = 36. Count up: 64 to 70 is 6, then 70 to 100 is 30, so 36. They match.

Take away: 100 - 60 = 40, and 40 - 4 = 36. Count up: 64 to 70 is 6, and 70 to 100 is 30, so 6 + 30 = 36. Both ways give 36. Check: 64 + 36 = 100.

Watch for: Writing 164 (adding).; Writing 40 (forgetting to take away the 4 ones).

G3. Mei says 83 - 47 and 86 - 50 have the same answer. Is she right? Find both answers and say why.

Answer: Yes. Both are 36. Both numbers go up by 3, so the gap does not change.

83 - 47: 47 to 50 is 3, 50 to 80 is 30, 80 to 83 is 3, so 36. 86 - 50 = 36. 83 became 86 and 47 became 50: both went up by 3, so the gap between them stayed 36.

Watch for: Finding only one answer.; Saying the answers differ because the numbers differ.

Investigation (page 5): When is counting up quicker?

Counting up is quicker when the two numbers are close together, so the gap is short (52 - 49 is 3). Taking away is quicker when the number taken away is small or a whole number of tens (52 - 10 is 42). If the numbers are far apart and the small number is awkward, both ways take longer.

Look for: A child who tries several pairs, says what they notice about how far apart the numbers are, and gives a reason. Accept any clear rule that matches examples. A child who also checks by adding is doing well.

Exit ticket (page 5)

E1. Take away.

Answer: 34

58 take away 20 is 38, and 38 take away 4 is 34.

Watch for: Writing 82 (adding).; Writing 36 (took away 2 and 2).

E2. Count up from 57 to 62.

Answer: 5, then 5

57 to 60 is 3, and 60 to 62 is 2. That is 5. So 62 - 57 = 5.

Watch for: Writing only the first hop (3).; Writing 15 (62 - 47).

E3. To find 90 - 87, which is quicker: take away or count up? Write the answer.

Answer: 3. Counting up is quicker: 87 and 90 are close together.

87 to 90 is 3, so 90 - 87 = 3. Counting up is quick because the numbers are close.

Watch for: Taking away 87 from 90 one at a time.; Writing 13.

Sneak-peek puzzles (not in the workbook)

1. 11 stars.
2. 4 more. From 58 to 60 is 2, and from 60 to 62 is 2 more. 2 and 2 make 4.